

BIOLOGY 101 · RECITATION HOMEWORK

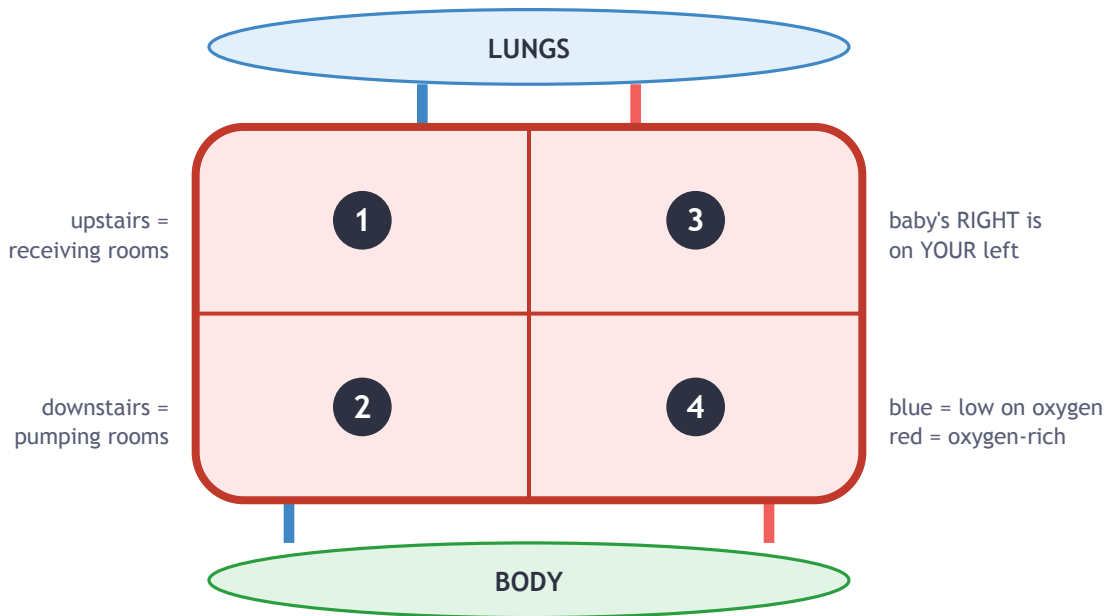
Homework 2: The Heart and the Blood

Name: _____

Date: _____

Instructions: About 20 minutes. Part A uses the diagram: remember the baby's right side is on YOUR left. Show your multiplication in Part B – the steps are worth points. Total: **24 points**.

Part A — Label the four rooms



1. For each number on the diagram, write the room's name and what it does (receives blood from where, or pumps blood to where). (8 pts)

#	Room	What it does
1		
2		
3		
4		

Part B — Harvey's arithmetic, on a newborn

2. A newborn's heart beats 130 times a minute. How many beats in an hour? In a day? Show the multiplication. (3 pts)

3. Count your own pulse for 15 seconds (two fingers on the side of your neck or the inside of your wrist). Multiply by 4. What's your heart rate? (1 pt)

Part C — What's in the blood

4. Draw a line from each part of the blood to its job. (4 pts)

Plasma	Plug leaks — the start of a scab
Red blood cells	Hunt and fight germs
White blood cells	The water that carries salt, sugar, and bicarbonate
Platelets	Carry oxygen with hemoglobin

Part D — Counting drops

5. A preemie weighs 1.5 kg. About how much blood does she have (80 mL per kg)? A blood test takes 3 mL. What fraction of her blood is that? (Hint: how many 3s fit into her total?) (2 pts)

Part E — Flashbacks

6. From NICU Tutor: the umbilical cord has three blood vessels. What are they, and what does a doctor use them for in the first days of life? (2 pts)

7. From this morning's Chemistry: the bicarbonate 'sponge' rides in which part of the blood? (1 pt)

Part F — Think like a doctor

8. A 900-gram preemie is five days old and breathing harder today than yesterday. The nurse hears a whooshing murmur. An ultrasound of the heart shows the ductus arteriosus is still wide open. Explain what is happening to the blood — and name two things the team might decide to do. (3 pts)

HOME LAB (optional, not graded): One-way doors in your own arm

You need: your arm, a mirror or a grown-up to watch, and a minute of patience

1. Let one arm hang straight down for a minute until the veins on the back of that hand puff up.
2. Press one puffy vein with a fingertip and keep pressing. With a finger of your other hand, stroke along the vein AWAY from your heart, toward your knuckles.
3. Look at the stretch you just emptied. Does it stay flat, or refill? Now lift the first finger.
4. Raise your hand above your head and watch the veins. Then hang it down again and watch them come back.
5. Bonus: count your pulse for 15 seconds, do 20 jumping jacks, and count again.

What I saw:

Done! Professor's initials: _____

Score: _____ / 24 points

Professor's comment:
